

Problem Statement ID - SIH26102

Problem Statement Title - Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Scheme implementation regd.

Theme - Smart Automation

PS Category - Software

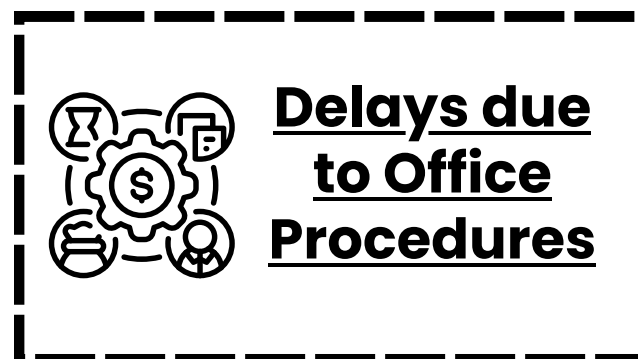
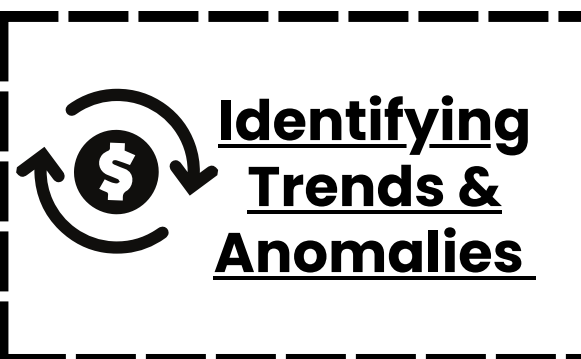
Team ID - 160236

Team Name - VIKSIT_YASH

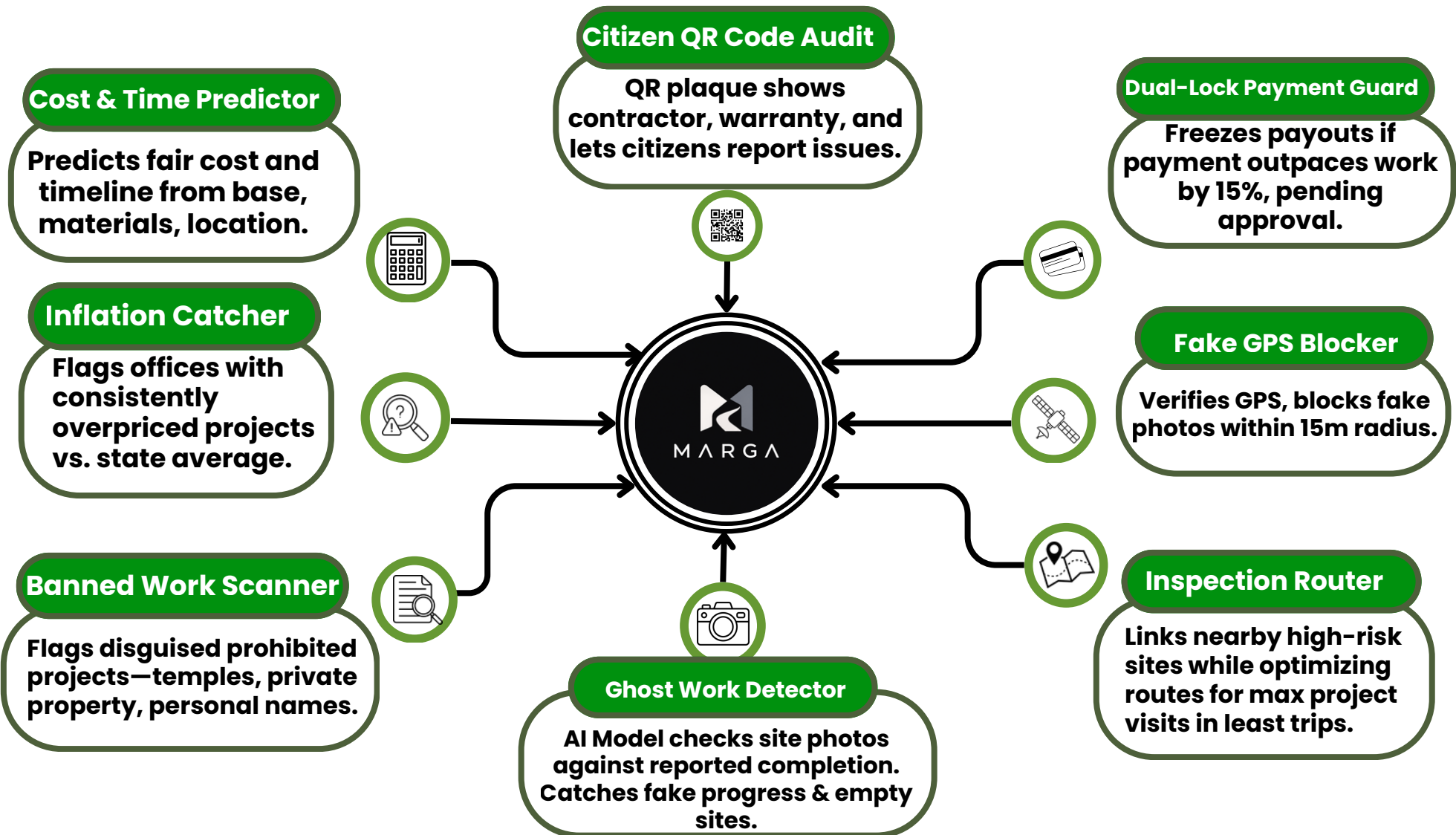


The Real World Problems

[News Papers, Articles Attached]



Our Innovation



Problems

- 1.PRE SANCTION
- 2.EXECUTION
- 3.COMPLETION
- 4.INSPECTION
- 5.OVERSIGHT

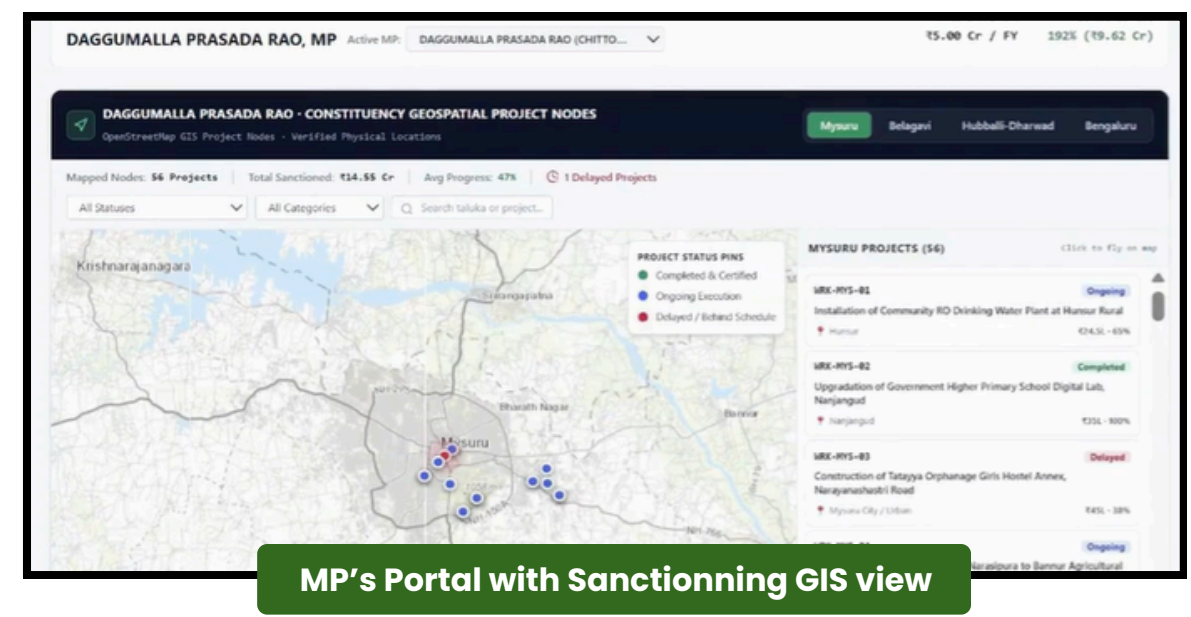
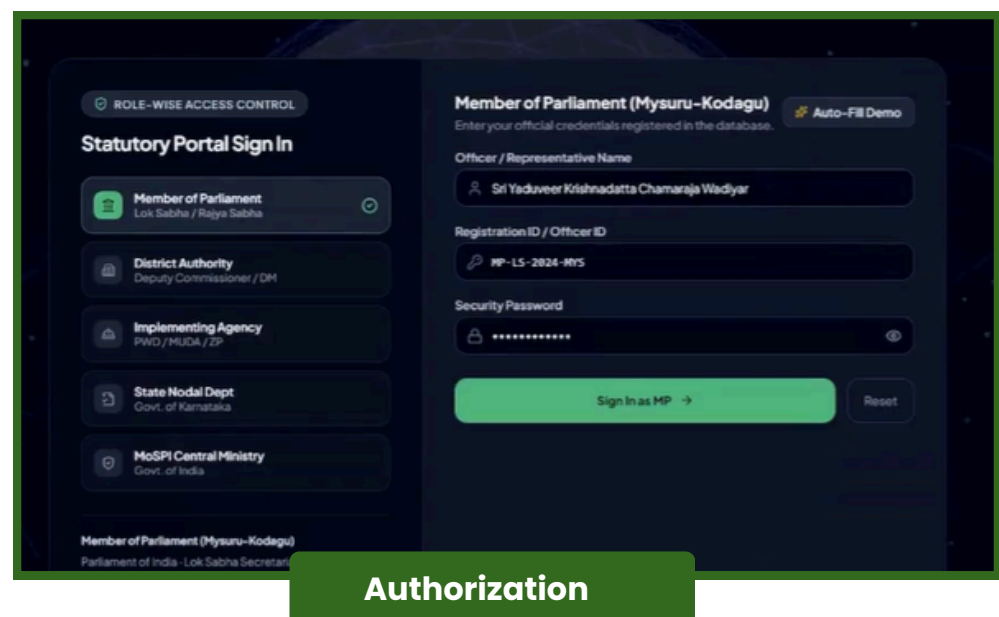
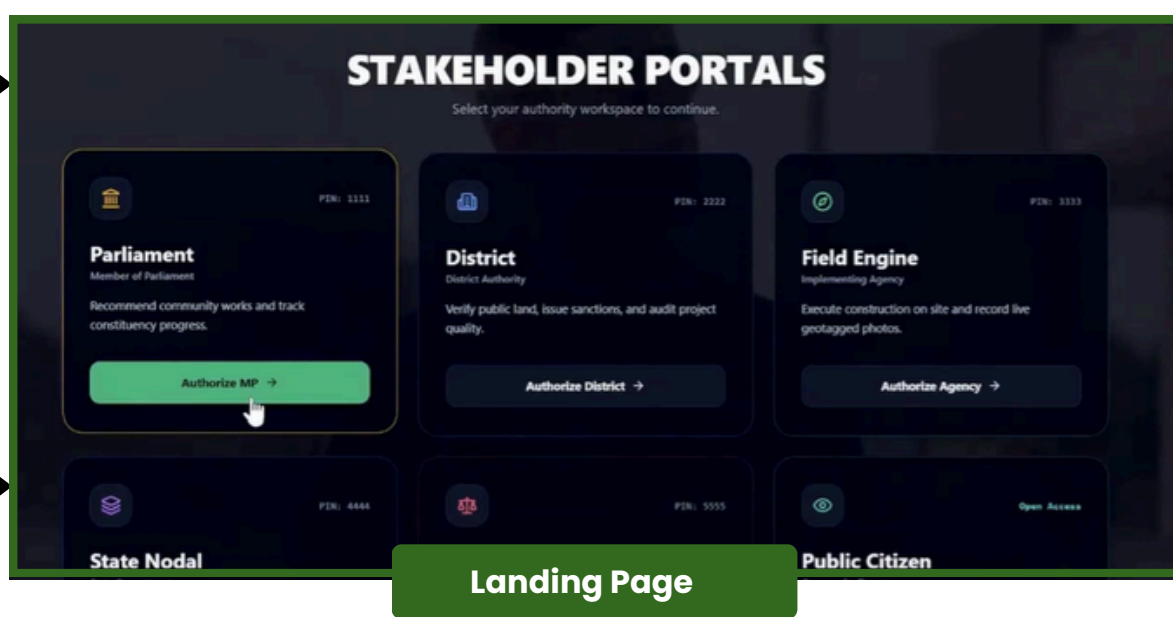
MARGA's Solutions

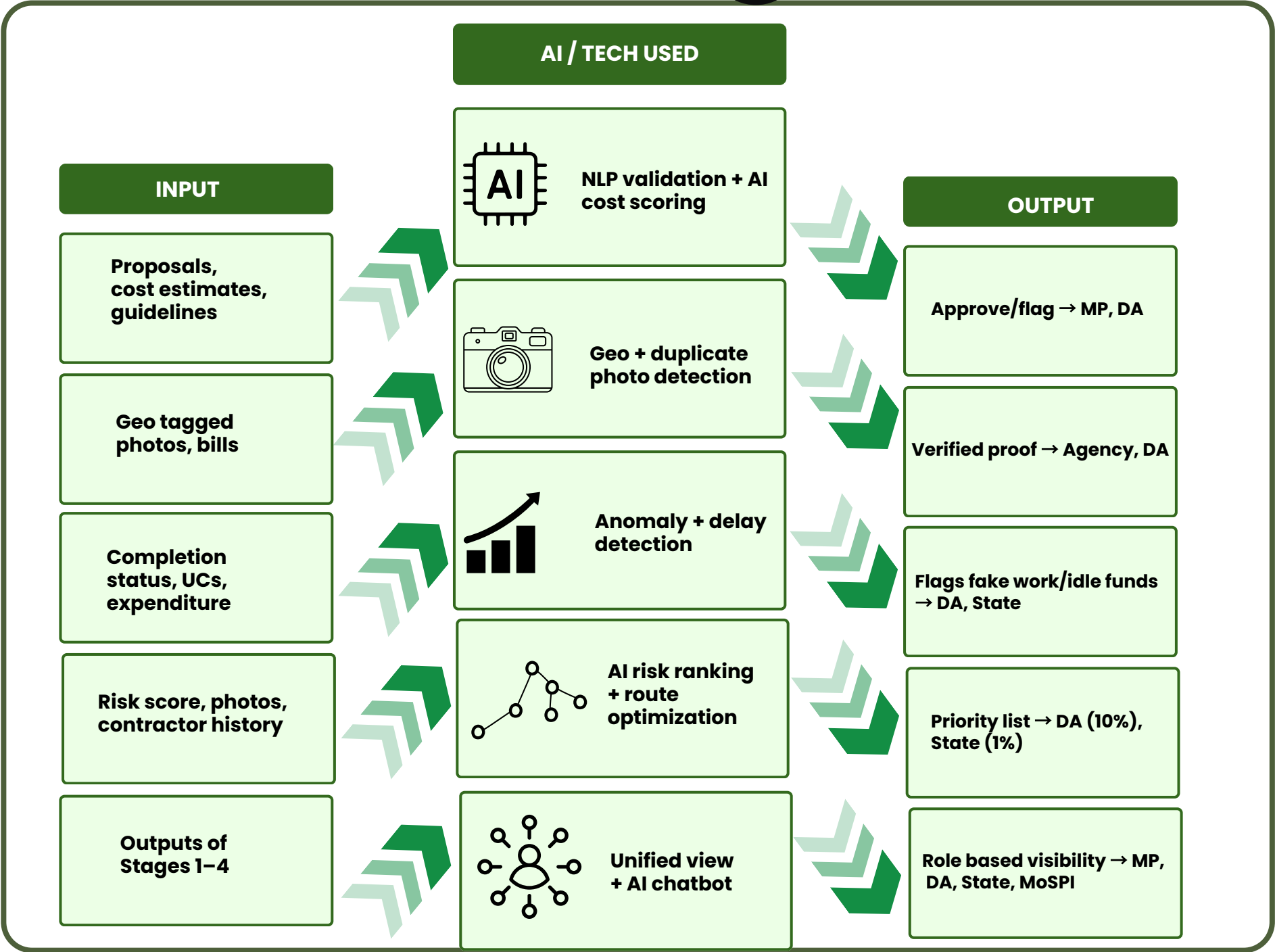
- Smart Proposal Checker
- Marga Eyes (Field App)
- Fund & Milestone Checker
- Smart Inspection Planner
- Dashboard+ Ask MARGA

What is Marga?

A low power, AI driven risk detection and decision support platform for government scheme oversight validating sanctions, suggesting field visits, reconciling funds, and routing inspections in real time, for smarter, fraud resistant public spending.

Web Portal



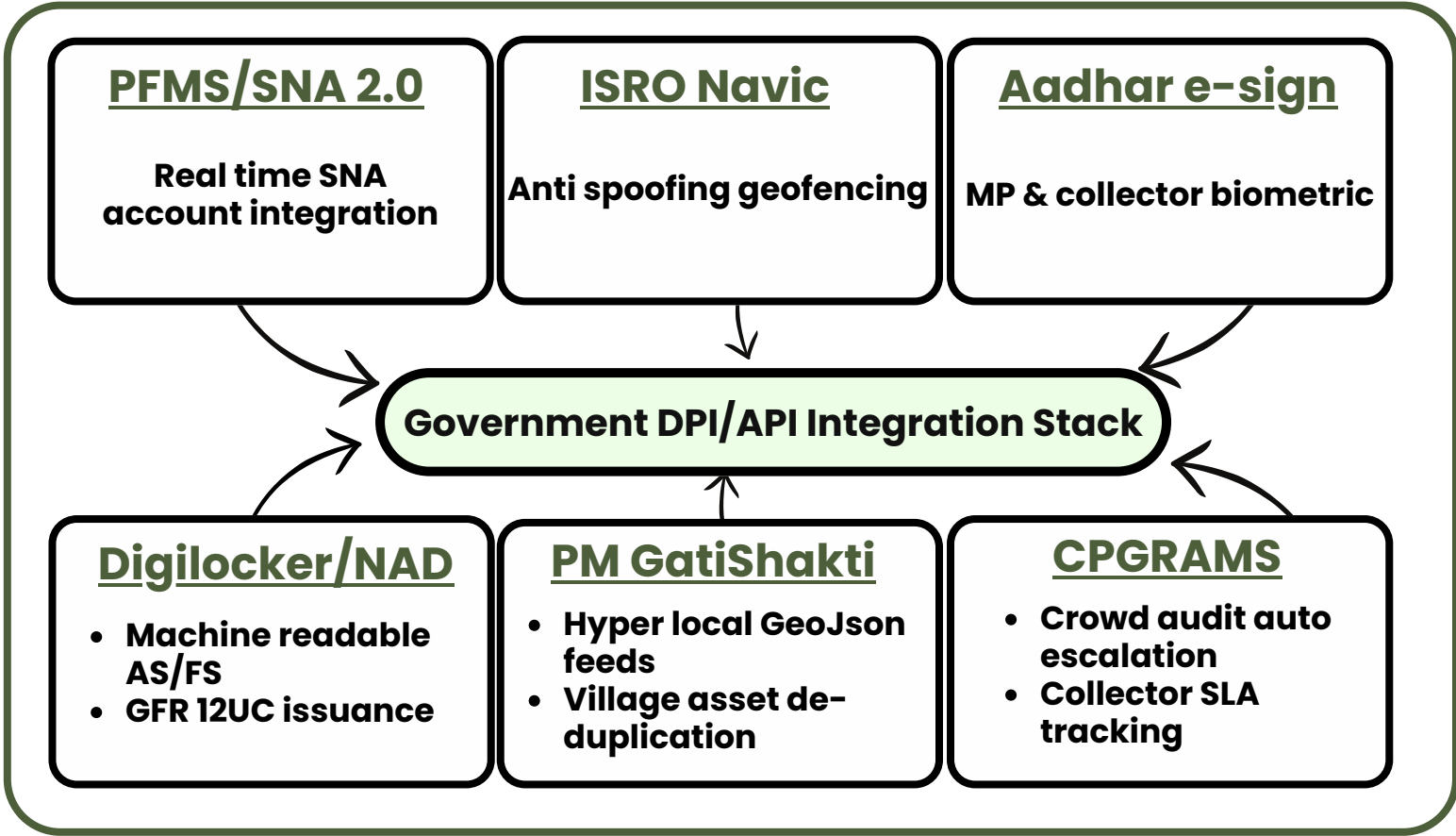


MARGA closes the MPLADS loop catching anomalies at every stage, from proposal to oversight, so funds reach real work, not fake records.

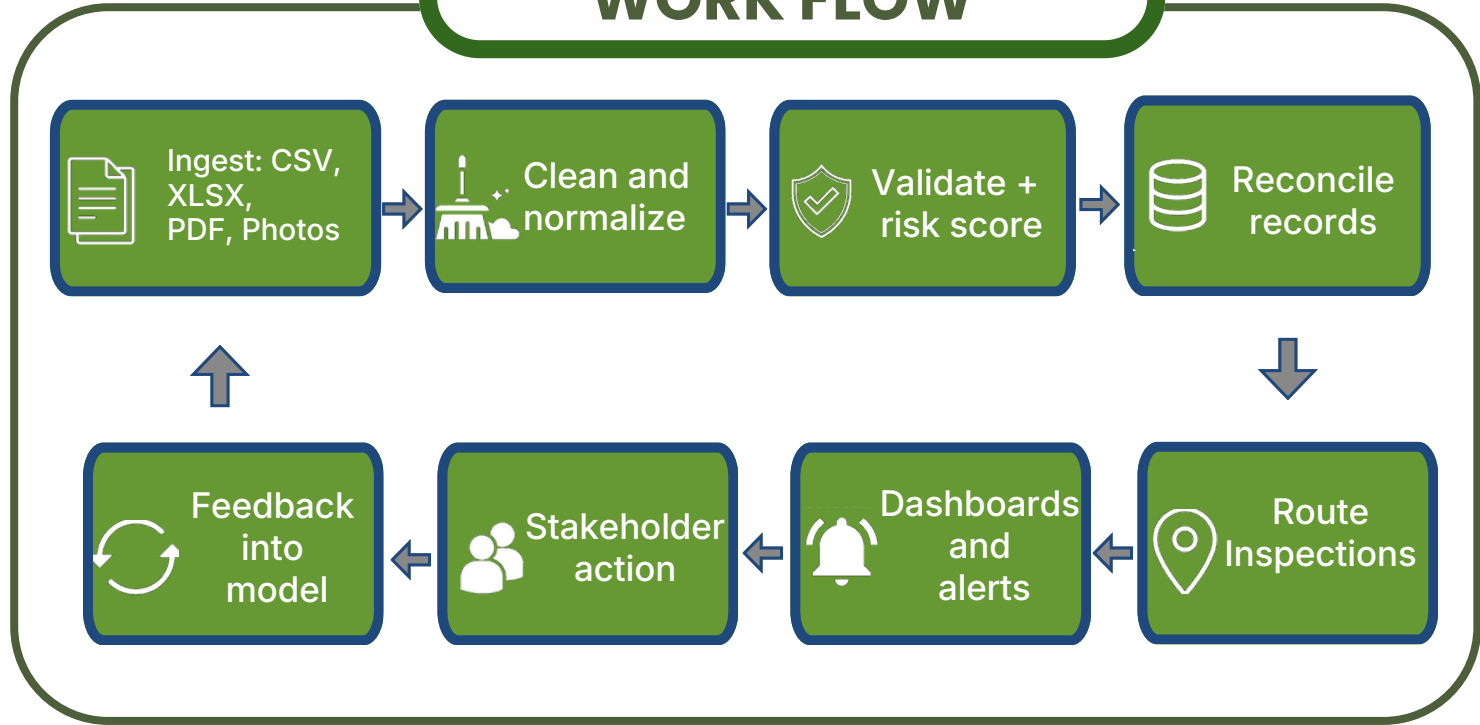
TECH STACK



PyTorch: deep learning models for photo and stage verification
Scikit learn: anomaly detection to flag unusual patterns
LightGBM: benchmarks costs to catch inflated billing
Bhashini: alerts in the officer's own language, voice too
WhatsApp: delivers alerts into officers' daily workflow
React: powers role based dashboards for all stakeholders



WORK FLOW



Minimum Government, Maximum Governance

From Feasibility to Reality

Challenges	Mitigation
False alarms → ignored alerts	alert only at Z > 3 over an agency's last 50 works
Evidence tampering	Every flag, photo and officer action is SHA-256 hash chained. Editing or deleting any past record breaks the chain and is detected.
Fake photos, weak network	geo + time stamp, duplicate check, offline sync. 90 idle days → show cause notice.
Policy approval, adoption	advisory mode first, read only eSAKSHI feed. Hard locks only after MoSPI/State approval.

FEASIBILITY ANALYSIS

Live from Day 1

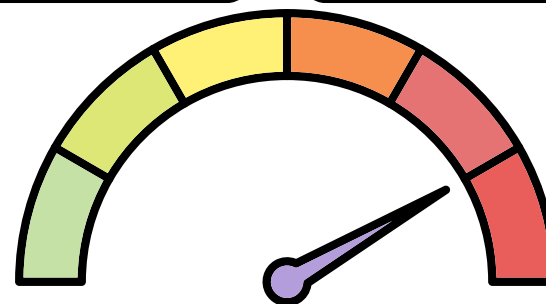
Rules + NLP check 15+ Annexure II clauses with 0 training records. ML joins as data grows.

Field ready in 3 taps

Photo, status, amount spent. About 100 photos per district a month (≈3 a day), ~300 KB each.

Real time and lightweight

Tree models score a work in <10 ms on ordinary CPU servers. No GPU needed for cost and NLP checks.



No rule changes needed

Automates existing mandates: 10% DA and 1% State inspections, UC, 90 day sanction window.

Why MARGA Is Viable

- Seconds-long cost card vs. 90-day manual estimate.
- 15 routes instead of 40 for the same inspection quota.
- No more backlog: UCs auto-generated at completion.
- Near-zero cost: ~₹7,200/year nationwide vs. crores lost to leakage.
- Automates mandates that already exist.

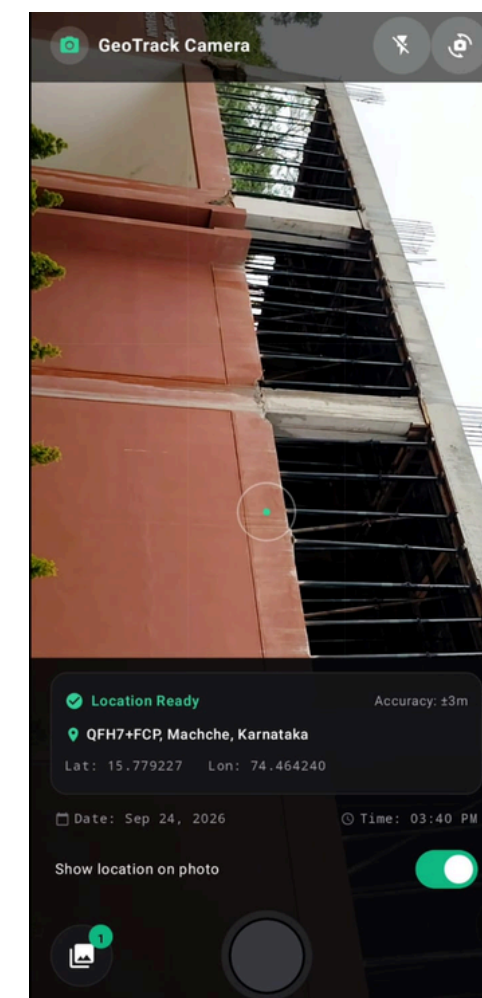
Saves officer time

Tiny running cost

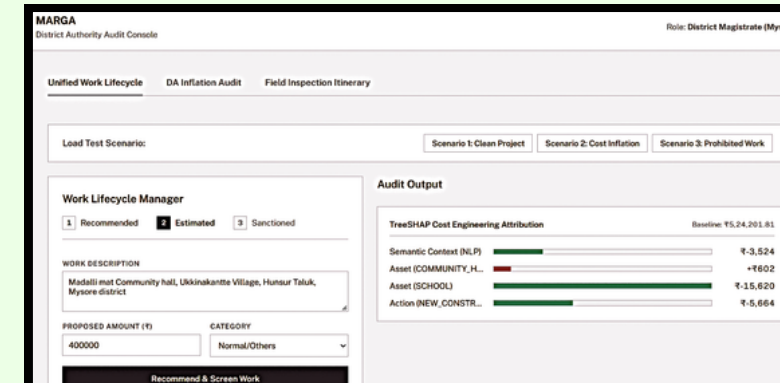
WOW FACTORS

More checks, less driving

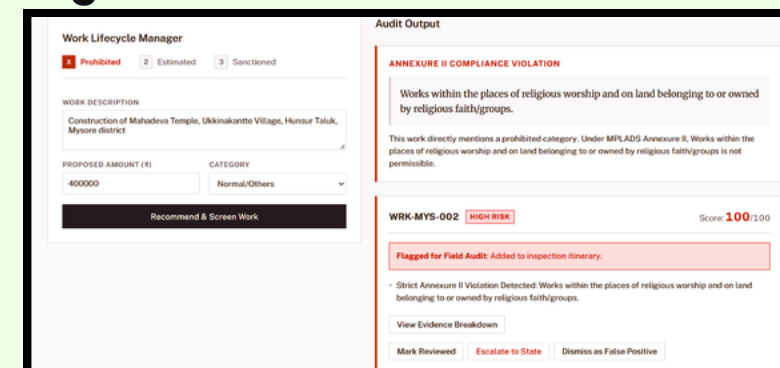
Ends the paperwork backlog



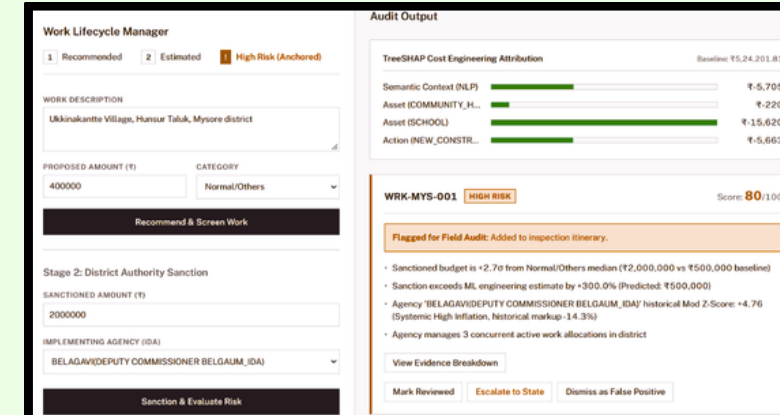
Mobile Application to detect Ghost work



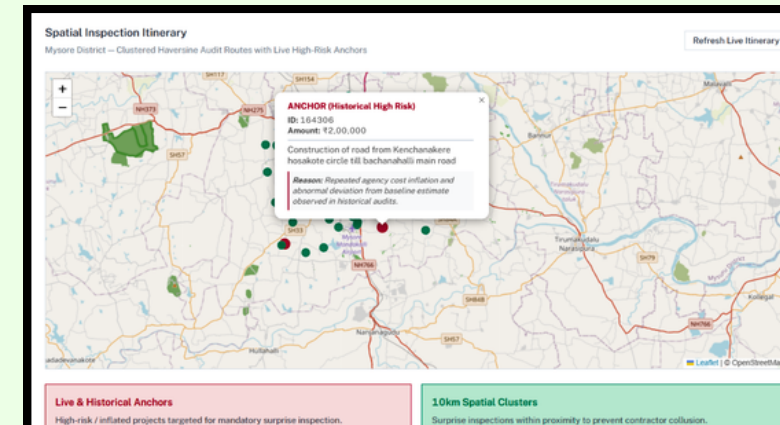
LightGBM Cost & timeline Estimator



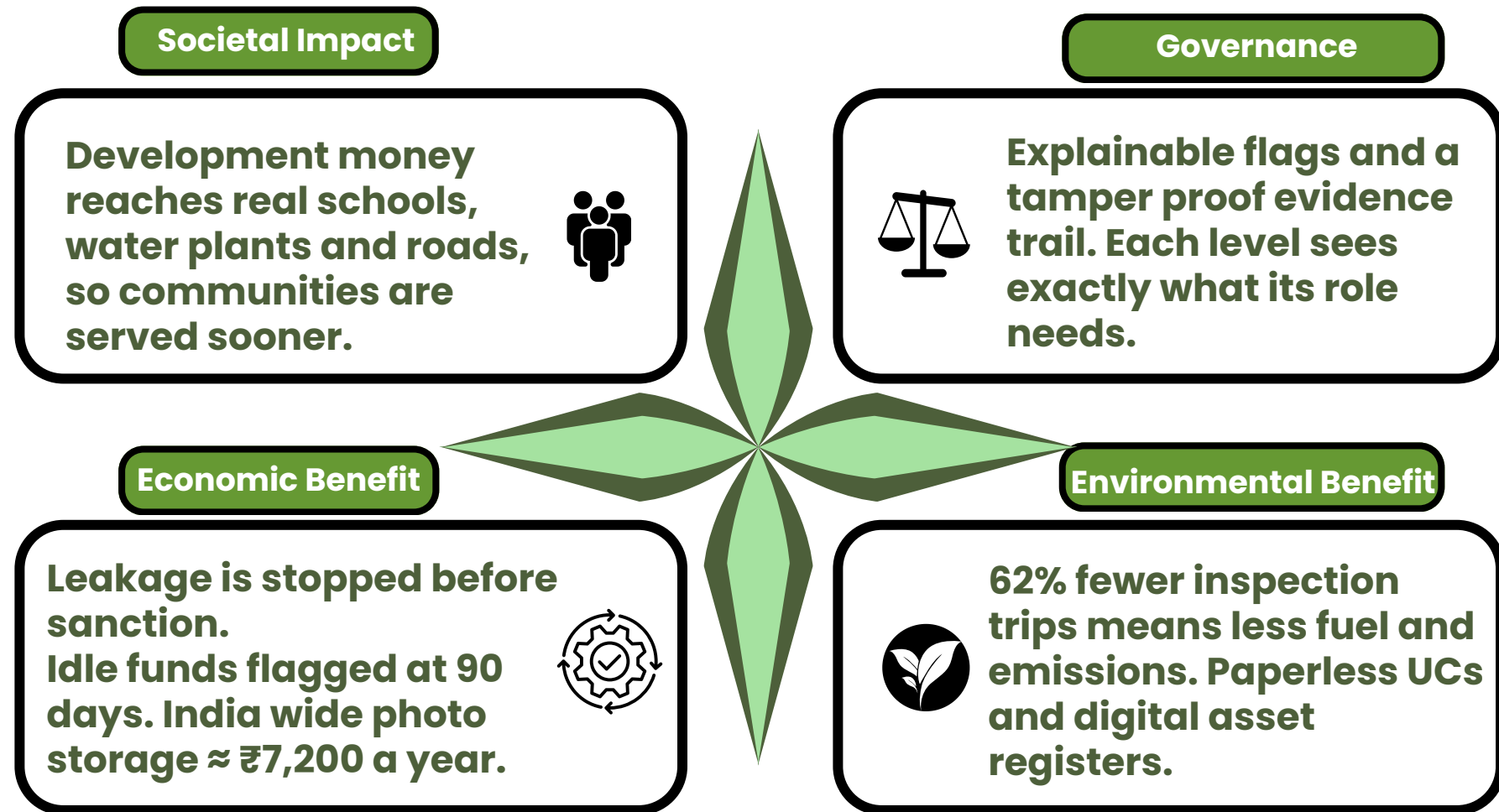
NLP based Prohibited work detector



Inflated Cost detector

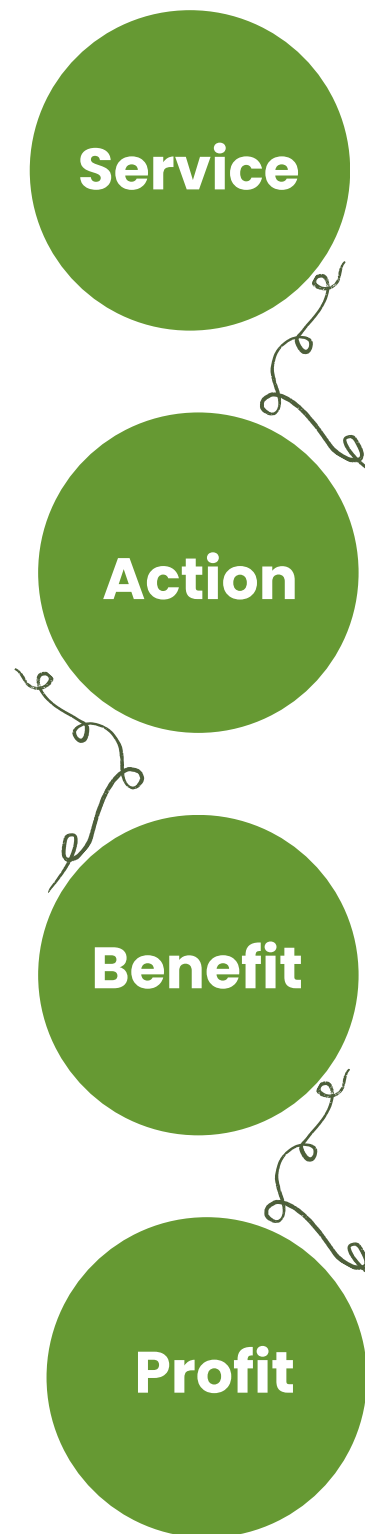


Optimised inspection route



"Risk Detected, Trust Protected."

- With AI driven validation, District Authorities catch inflated or prohibited proposals before sanction, not after the damage is done.
- With geo tagged field evidence, ghost assets and duplicate billing are exposed in real time, not years later in an audit.



DIRECT IMPACT ON TARGET USER

- **MP:** Plain language "Needs Attention" list + ready-to-send letter to DA — before damage is done.
- **District Authority:** Cost card in <10ms, auto compiled reports, route planned inspections.
- **Implementing Agency:** 3 tap updates, auto UC paperwork, Trust Score for on time delivery.
- **State Nodal Authority:** Risk heat map + conflict based inspection routes 5 works/2 day drive.
- **MoSPI & CAG:** Data Integrity Score alongside UC compliance to catch ghost assets paper hides.
- **Citizens:** Real, geo tagged assets not just paper claims.

Impact Proofs

Verification gap closed 27.3% $(3,913 - 2,843) \div 3,913$ CAG audit, Andhra Pradesh: 3,913 works reported complete, only 2,843 physically verified. MARGA's Reconciliation Engine blocks this gap from forming.	Inspection trips saved 62.5% $(40 - 15) \div 40$ Random 10% quota needs 40 scattered trips across a 13,000 sq km district. Anchor-based spatial clustering hits the same quota in 15.	National audit-trail cost 500rs/mo National audit-trail storage: ~₹500/month/year¹ (270GB, compressed geo-photos, ₹1.9/GB) ¹cost scales with cumulative photos; cold-storage archival after 1 year keeps this flat
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GOVERNMENT & POLICY

1 MPLADS Guidelines 2023

WHY WE USE IT

The rulebook: prohibited works, 10% and 1% inspections, and UC rules.

Used in: Rule engines

mplads.gov.in

2 MPLADS-eSAKSHI portal

WHY WE USE IT

The live system MARGA reads from: works, sanctions, status, spending.

Used in: Data Input

mplads.mospi.gov.in

3 National Strategy for AI

WHY WE USE IT

National policy push for responsible, transparent AI in public services.

Used in: Explainable Design

niti.gov.in · [AI strategy \(PDF\)](#)

4 CAG Report No. 31 of 2010-11

WHY WE USE IT

Auditors' real findings: prohibited works, delays, doubtful spending.

Used in: Problem & test cases

cag.gov.in · [Report 31](#)

MACHINE LEARNING & ANOMALY DETECTION

5 LightGBM: A Highly Efficient Gradient Boosting Decision Tree

WHY WE USE IT

Fast, accurate cost prediction; trains up to 20× faster at near equal accuracy.

Used in: Cost Scoring

[NeurIPS proceedings \(PDF\)](#)

6 A Unified Approach to Interpreting Model Predictions (SHAP)

WHY WE USE IT

Shows why a cost was flagged, feature by feature, so officers can trust it.

Used in: TreeSHAP explanations

arxiv.org/abs/1705.07874

7 scikit-learn: BallTree

WHY WE USE IT

Fast nearest site search on map coordinates to build short inspection routes.

Used in: Inspection Routes

scikit-learn.org · [BallTree](#)



Our aim is to build an ecosystem every officer can use.

MARGA provides simple dashboards, an easy geotagging app, and clear risk alerts in the officer's own language through familiar tools like WhatsApp. This helps even **non technical** officers perform their duties efficiently and confidently.

DETAILED REPORT: [GITHUB](#)



DISTRICT AUTHORITY VALIDATION

Idea shared with a District Authority official.
Email received from DPO Kolhapur on
3 Sept 2026.